1. Who is known as the father of AI and why?

John McCarthy is known as the father of AI because he coined the term “Artificial Intelligence” in 1956 and organized the Dartmouth Conference. He developed the LISP language, which became the primary AI programming language. His contributions to symbolic reasoning, time-sharing systems, and AI theory built the foundation for the field.

2. Discuss the importance and challenges of Natural Language Processing (NLP) and Computer Vision in AI.

NLP and Computer Vision are two major AI fields that let machines understand human language and visual data respectively. NLP enables applications like chatbots, translation, and search by converting text/speech into structured meaning, while Computer Vision powers object detection, image recognition, and autonomous driving by extracting features from images and video. Their importance lies in enabling natural human–machine interaction and automating perception-heavy tasks. Challenges include large variability (languages, dialects, image conditions), the need for huge annotated datasets, ambiguity (word senses, occluded objects), fairness and bias in data, and high compute/resources for state-of-the-art models. Both fields also face deployment issues such as real-time constraints, adversarial attacks, and domain adaptation when models move from lab data to real-world conditions.

3. What is meant by percept in intelligent agents?

A percept is the raw input an intelligent agent receives from its sensors at a single time step. It could be an image frame for a vision agent, a sound snippet for a speech agent, or a sentence token for an NLP agent. The agent’s action decision is based on the percept sequence (its percept history) and its internal state. In short, percepts are the building blocks of the agent’s view of the world and determine how it updates beliefs and chooses actions.

4. Write a brief note on the foundations and history of AI.

AI’s foundations combine logic, probability, learning, and optimization, with early roots in symbolic logic, search, and knowledge representation. The field formally began in the 1950s with pioneers like Turing, McCarthy, Newell, and Simon; early work focused on rule-based systems and search. Progress slowed during the “AI winters” when expectations outpaced results. The revival began with statistical learning, better algorithms, larger datasets, and faster hardware, leading to modern machine learning and deep learning breakthroughs in the 2010s. Today’s AI integrates symbolic reasoning, probabilistic models, and neural networks, applied across vision, language, robotics, and more.

5. Describe in detail the types of intelligent agents with suitable examples.

Intelligent agents vary by their design and capabilities. Simple reflex agents act only on current percepts using condition-action rules (e.g., a thermostat). Model-based reflex agents keep internal state to handle partially observable worlds (e.g., a navigation controller that tracks position). Goal-based agents choose actions to achieve explicit goals (e.g., a path-planning robot). Utility-based agents optimize a utility function for trade-offs (e.g., a recommendation system balancing relevance and diversity). Learning agents improve performance from experience (e.g., a spam filter adapting to new spam types). Hybrid agents combine these aspects to tackle complex real-world tasks.

6. Explain in detail the concept of intelligent agents and their types with suitable examples.

An intelligent agent perceives its environment through sensors and acts through actuators to achieve objectives. Key components are percepts, a performance measure, an environment, and an agent program that maps percepts to actions. Agents are classified by autonomy and capability: reflex (thermostat), model-based (Roomba vacuum that maps rooms), goal-based (chess engine aiming to checkmate), utility-based (stock trading agent maximizing expected return), and learning agents (language models improving via fine-tuning). In practice, most systems are hybrid: a self-driving car uses perception (vision), planning (goal/utility), and learning components together.

7. Define Artificial Intelligence.

Artificial Intelligence is the branch of computer science that studies creation of systems able to perform tasks that normally require human intelligence. These tasks include learning from data, reasoning, planning, natural language understanding, perception, and decision-making. AI systems range from narrow AI, built for specific tasks like translation, to the research goal of general intelligence that can adapt across many tasks.

8. Explain statistical pattern recognition system in detail.

Statistical pattern recognition uses probability and statistics to classify or recognize patterns from features extracted from data. The system pipeline includes feature extraction, modeling class-conditional densities (e.g., Gaussian models), estimating parameters from training data, and using decision rules (like Bayes decision rule) to assign labels. It handles noise and variability by modeling uncertainty; for example, in digit recognition, pixel features feed into probabilistic models or classifiers (Naive Bayes, GMMs). Challenges include selecting discriminative features, model selection, and avoiding overfitting, but statistical methods provide principled performance guarantees and probabilistic confidence.

9. Define Natural Language Processing.

Natural Language Processing is the field that enables computers to understand, interpret, generate, and respond to human language in text or speech form. It combines linguistics and machine learning to solve problems like tokenization, POS tagging, parsing, semantic understanding, machine translation, summarization, and dialogue. Modern NLP relies heavily on statistical and neural methods (e.g., transformers) trained on large corpora to capture syntax and semantics..

10. Explain how heuristics improve search efficiency.

Heuristics are informed rules or estimates that guide search toward promising states, reducing the number of nodes expanded compared to blind search. A good heuristic approximates the cost-to-go (e.g., straight-line distance in pathfinding) and helps algorithms like A* prioritize states that likely lead to the goal faster. Heuristics can make exponential search spaces tractable by pruning unlikely branches and focusing computation, while admissible heuristics guarantee optimal solutions when used with A*. In practice, domain knowledge encoded as heuristics dramatically speeds up planning and puzzle solving.

11. What is the Turing Test?

The Turing Test, proposed by Alan Turing, is a behavioral test of machine intelligence: if a human interrogator conversing through a text channel cannot reliably distinguish the machine from a human, the machine is said to exhibit intelligent behavior. It emphasizes indistinguishable conversational performance rather than internal mechanisms. While influential, the Turing Test has limitations: it measures imitation, can be gamed, and does not capture reasoning depth or other cognitive capabilities.

12. List and explain any three important applications of AI in daily life.

Artificial Intelligence is widely used in daily life applications to make tasks easier and more efficient. One common application is **virtual assistants** like Siri, Alexa, and Google Assistant, which understand voice commands and help in tasks like setting reminders and searching information. Another application is **recommendation systems** used by YouTube, Netflix, and Amazon, which analyze user behavior to suggest relevant videos or products. AI is also used in **smartphone cameras** for face recognition, image enhancement, and night mode photography. These applications show how AI improves convenience, personalization, and user experience in everyday activities.

13. What are Computer Vision and Natural Language Processing (NLP)? Explain their working and applications with examples.

Computer Vision enables machines to understand visual data such as images and videos, while NLP helps computers understand and process human language. Computer Vision works by capturing images, extracting features, and classifying objects using machine learning models; for example, face recognition in mobile phones. NLP works by processing text through steps like tokenization, parsing, and semantic analysis; for example, chatbots answering user questions. Applications of Computer Vision include medical image analysis and autonomous vehicles, whereas NLP is used in machine translation, sentiment analysis, and voice assistants.

14. Discuss local search algorithms and their role in optimization.

Local search algorithms focus on improving a single current solution rather than exploring entire state spaces. They work by moving from one solution to a neighboring better solution based on an evaluation function. Algorithms like **hill climbing** and **simulated annealing** are commonly used. Local search plays an important role in optimization problems such as scheduling, routing, and resource allocation because they require less memory and computation. Although they may get stuck in local optima, they are effective for large and complex problems.

15. Explain statistical learning theory with models.

Statistical learning theory provides a mathematical framework to understand how machines learn patterns from data and generalize to unseen examples. It focuses on concepts like hypothesis space, training error, generalization error, and model complexity. Models such as linear classifiers, decision trees, and support vector machines are studied under this theory. The goal is to balance underfitting and overfitting by selecting models that perform well on both training and test data. It helps in designing reliable and efficient learning algorithms.

16. Define a knowledge base.

A knowledge base is a structured collection of facts, rules, and relationships about a particular domain that an AI system uses for reasoning and decision-making. It stores information in formal representations such as propositional logic, first-order logic, or frames. Expert systems rely heavily on knowledge bases to simulate human expertise. For example, a medical diagnosis system stores symptoms and disease rules in a knowledge base to suggest possible diagnoses.

17. Explain how reasoning techniques are used in autonomous systems to make decisions when information is incomplete.

Autonomous systems often operate in environments where complete information is not available. Reasoning techniques such as probabilistic reasoning, Bayesian inference, and fuzzy logic help these systems make decisions under uncertainty. For example, a self-driving car uses sensor data with noise and uncertainty to predict obstacles and select safe actions. These techniques allow the system to estimate probabilities, update beliefs, and choose actions that maximize expected outcomes despite incomplete data.

18. Define Hidden Markov Model (HMM).

A Hidden Markov Model is a probabilistic model used to represent systems where the states are hidden but produce observable outputs. It consists of hidden states, observable symbols, state transition probabilities, emission probabilities, and initial state probabilities. HMMs are widely used in speech recognition, handwriting recognition, and bioinformatics. For example, in speech recognition, the spoken sound is observable, but the linguistic state generating it is hidden.

19. Explain how probabilistic reasoning improves spam email classification.

Probabilistic reasoning improves spam detection by handling uncertainty and variations in email content. Techniques like **Naive Bayes classifiers** calculate the probability of an email being spam based on the frequency of certain words. The system learns from past labeled emails and updates probability estimates as new data arrives. This approach is effective because it can adapt to new spam patterns and reduces false classification by using likelihood-based decisions.

20 Define supervised learning.

Supervised learning is a type of machine learning where a model is trained using labeled data, meaning each input has a known output. The algorithm learns a mapping from inputs to outputs by minimizing errors on training examples. Common supervised learning algorithms include linear regression, decision trees, and support vector machines. It is widely used in applications like spam detection, image classification, and disease prediction.

21 Explain forward chaining with an example?

Forward chaining is a data-driven reasoning method used in rule-based systems where inference starts from known facts and applies rules to derive new facts until a goal is reached. For example, if a system knows that “If it rains, roads are wet” and it is given the fact “It is raining,” it can conclude that “The roads are wet.” Forward chaining is commonly used in expert systems where conclusions are derived step by step from available information.

22 Define the working of minimax algorithm?

The minimax algorithm is used in two-player games to choose the best possible move assuming that the opponent also plays optimally. It works by generating a game tree and assigning values to terminal states as win, loss, or draw. The maximizing player selects the move with the highest value, while the minimizing player selects the move with the lowest value. This process continues alternately until the root of the tree is evaluated. Minimax ensures optimal decision-making in games like chess, tic-tac-toe, and checkers.

23 Explain adversarial search and its application in game playing.

Adversarial search is a search technique used in environments where multiple agents compete with opposite goals. Each agent tries to maximize its own success while minimizing the opponent's. Algorithms like minimax and alpha–beta pruning are used to analyze possible future moves. In game playing, adversarial search helps AI systems choose the best move by considering opponent responses. This approach is widely used in board games such as chess, Go, and strategy-based video games.

24 What is pattern recognition?

Pattern recognition is the process of identifying patterns or regularities in data using computational techniques. It involves collecting data, extracting features, and classifying or labeling the input based on learned patterns. Pattern recognition is used in applications such as image recognition, speech recognition, and handwriting analysis. The goal is to allow machines to automatically recognize similarities or differences in input data.

25 Explain the need for probabilistic reasoning.

Probabilistic reasoning is needed to make decisions in situations where information is uncertain or incomplete. Real-world environments often involve noise, ambiguity, and unpredictability. Probability theory allows AI systems to represent uncertainty and update beliefs based on new evidence. For example, medical diagnosis systems use probabilistic reasoning to estimate the likelihood of diseases based on symptoms. This approach improves decision accuracy and reliability under uncertainty.

26 What is Bayes classifier?

A Bayes classifier is a probabilistic classifier based on Bayes' theorem, which calculates the probability of a class given observed features. The most common example is the Naive

Bayes classifier, which assumes independence among features. It is simple, fast, and effective for large datasets. Bayes classifiers are widely used in spam filtering, document classification, and sentiment analysis.

27 Discuss challenges and limitations of machine learning.

Machine learning faces several challenges such as the need for large amounts of high-quality data and high computational cost. Models may suffer from overfitting, where performance is good on training data but poor on new data. Bias in datasets can lead to unfair or inaccurate predictions. Machine learning models are often difficult to interpret, making decision explanation challenging. These limitations must be addressed to build reliable AI systems.

28 Write short notes on SVM.

Support Vector Machine (SVM) is a supervised learning algorithm used for classification and regression tasks. It works by finding an optimal hyperplane that separates classes with maximum margin. SVM can handle non-linear data using kernel functions such as linear, polynomial, and radial basis functions. It is effective in high-dimensional spaces and commonly used in text classification and image recognition.

29 Explain knowledge representation using propositional logic and its limitations.

Propositional logic represents knowledge using propositions that are either true or false, connected by logical operators such as AND, OR, and NOT. It is simple and useful for basic reasoning tasks in AI systems. However, it cannot express relationships between objects or handle variables and quantifiers. Due to this limitation, propositional logic is not suitable for complex real-world knowledge representation.

30 Explain role of dimensionality reduction in pattern recognition.

Dimensionality reduction reduces the number of features in a dataset while preserving important information. It improves classification accuracy, reduces noise, and lowers computational cost. Techniques such as PCA and LDA are commonly used. In pattern recognition, dimensionality reduction helps remove redundant features and avoids the curse of dimensionality, making models faster and more efficient.

31 Explain how forward chaining is used in expert systems for agriculture.

In agriculture expert systems, forward chaining starts with collected facts such as soil moisture, temperature, and crop type. Rules are applied to infer conditions like pest risk or fertilizer requirements. For example, if soil moisture is low and temperature is high, the system may infer the need for irrigation. Forward chaining helps farmers make timely decisions based on real-time data.

32 Discuss types of features used in pattern recognition.

Features in pattern recognition are measurable properties used to represent data effectively. Common types include **statistical features** like mean, variance, and texture measures, which describe overall data behavior. **Structural features** represent shape and structure, such as edges, corners, and contours in images. **Transform-based features** are obtained using transformations like Fourier or Wavelet transforms to capture frequency information. Selecting appropriate features is important because they directly affect recognition accuracy and system performance.

33 Define inference in logic?

Inference in logic is the process of deriving new facts or conclusions from known facts and rules. It allows AI systems to reason and make decisions based on logical relationships. Inference methods include deduction, induction, and abduction. For example, if “All humans are mortal” and “Socrates is human” are known, inference allows concluding that “Socrates is mortal.” Inference is essential in expert systems and reasoning engines.

34 What is hill climbing?

Hill climbing is a local search algorithm used for optimization problems. It works by starting with an initial solution and repeatedly moving to a neighboring solution that improves the objective function value. The process stops when no better neighboring state is found. Hill climbing is simple and memory-efficient but can get stuck in local maxima or plateaus. It is commonly used in scheduling and optimization tasks.

35 Define the term State Space?

State space is the complete set of all possible states that a problem-solving agent can reach from the initial state by applying available actions. Each state represents a unique configuration of the problem. In AI search problems, algorithms explore the state space to find a path from the initial state to the goal state. For example, in the 8-puzzle problem, each board arrangement is a state.

36 Construct a truth table for the formula: (P→Q)∧(¬Q∨R). Determine whether the formula is satisfiable, valid or unsatifiable.

A truth table for the formula

$$(P \rightarrow Q) \wedge (\neg Q \vee R)$$

$(P \rightarrow Q) \wedge (\neg Q \vee R)$

is constructed below

P	Q	R	P → Q	¬Q	¬Q ∨ R	(P → Q) ∧ (¬Q ∨ R)
T	T	T	T	F	T	T
T	T	F	T	F	F	F
T	F	T	F	T	T	F
T	F	F	F	T	T	F
F	T	T	T	F	T	T
F	T	F	T	F	F	F
F	F	T	T	T	T	T
F	F	F	T	T	T	T

37 Discuss the components of HMM.

A Hidden Markov Model consists of five main components: a set of hidden states, a set of observable symbols, state transition probabilities, observation emission probabilities, and initial state probabilities. The hidden states represent underlying processes, while observations are visible outputs. HMM models sequences where the system state is not directly observable. These components allow HMMs to model time-dependent and uncertain processes like speech or gesture recognition.

38 Describe training, validation, and testing processes in ML.

In machine learning, data is divided into three parts to build reliable models. The **training set** is used to learn model parameters. The **validation set** helps tune hyperparameters and prevent overfitting. The **testing set** evaluates final model performance on unseen data. This separation ensures that the model generalizes well rather than memorizing data.

39 Explain adversarial search in AI.

Adversarial search is used in competitive environments where one agent's success is another's failure. It assumes rational opponents and evaluates moves by considering possible counter-moves. Algorithms like minimax and alpha–beta pruning are used to compute optimal strategies. Adversarial search is commonly applied in games such as chess, checkers, and Go.

40 Explain propositional logic with an example.

Propositional logic is a formal system where statements are represented as propositions that are either true or false. Logical connectives such as AND, OR, NOT, and IMPLIES are used to form complex expressions. For example, "If it rains, roads are wet" can be written as $P \rightarrow Q$. Propositional logic is simple and useful for basic reasoning tasks but limited in expressiveness.

41 Compare human intelligence and machine intelligence with examples.

Human intelligence is natural, flexible, and based on emotions, creativity, and consciousness. Humans can adapt easily and learn from very few examples. Machine intelligence is artificial and task-specific, relying on data, algorithms, and computation power. For example, humans understand language context naturally, while machines require large datasets and training. However, machines can process massive data faster and more accurately for specific tasks such as calculations or pattern detection.

42 Explain greedy best-first search.

Greedy best-first search is an informed search strategy that selects the next node to expand based only on a heuristic function. The heuristic estimates how close a node is to the goal, and the algorithm always chooses the node with the lowest heuristic value. It is fast and often finds a solution quickly, but it does not guarantee the optimal solution because it ignores the cost taken so far. Greedy best-first search is useful in problems where speed is more important than optimality, such as route finding and path planning.

43 Explain the difference between simple reflex and model-based agents.

A simple reflex agent makes decisions based only on the current percept, using condition–action rules. It does not remember past states, so it works well only in fully observable environments. In contrast, a model-based agent maintains an internal model of the world and uses it to keep track of unobserved aspects. For example, a simple reflex agent may turn on lights when it is dark, while a model-based agent can remember room layouts and previous sensor readings to make better decisions.

44 What is the role of sensors and actuators in an agent?

Sensors and actuators form the interface between an intelligent agent and its environment. Sensors collect percepts from the environment, such as cameras, microphones, or temperature sensors. Actuators allow the agent to perform actions, such as motors, speakers, or displays. Together, sensors provide input and actuators enable output, allowing the agent to perceive, decide, and act effectively in its environment.

45 Define optimal solution.

An optimal solution is the best possible solution among all feasible solutions according to a given performance measure. In AI search problems, an optimal solution usually means the least-cost path from the initial state to the goal state. Algorithms like A* with admissible heuristics are designed to find optimal solutions. Optimality ensures maximum efficiency and correctness in problem-solving.

46 Describe various informed search strategies with proper evaluation functions.

Informed search strategies use heuristic information to guide the search process. **Greedy best-first search** uses only a heuristic function h(n). **A*** search combines path cost and heuristic using the evaluation function f(n)=g(n)+h(n). **Hill climbing** uses a local evaluation function to move toward better neighbors. These strategies reduce search effort compared to uninformed search and are effective when good heuristics are available.

47 What is utility theory?

Utility theory provides a framework for making rational decisions by assigning numerical values to outcomes based on preferences. An intelligent agent selects actions that maximize expected utility. Utility theory is widely used in decision-making under uncertainty, economics, and game theory. In AI, it helps agents compare different outcomes and choose the most beneficial action.

48 Describe the role of Bayesian networks in probabilistic reasoning.

Bayesian networks are graphical models that represent variables and their probabilistic dependencies using a directed acyclic graph. They allow efficient reasoning under uncertainty by combining prior knowledge with observed evidence. Each node represents a random variable, and edges represent conditional dependencies. Bayesian networks are used in applications such as medical diagnosis, fault detection, and decision support systems.

49 Compare supervised, unsupervised, and reinforcement learning.

Supervised learning uses labeled data to learn a mapping from inputs to outputs, such as in classification problems. Unsupervised learning works on unlabeled data to discover patterns and structures, like clustering. Reinforcement learning involves an agent learning through interaction with the environment by receiving rewards or penalties. Each learning type suits different problem domains and data availability.

50 Mention any two tasks performed by intelligent agents.

Two common tasks performed by intelligent agents are **decision making** and **problem solving**. In decision making, the agent selects the best action based on percepts and goals, such as choosing a route. In problem solving, the agent searches for a sequence of actions to reach a goal state, such as solving puzzles or planning tasks.

51 Compare NN rule, Bayes classifier, and SVM.

Neural networks learn complex patterns using layered structures and are powerful for large and non-linear datasets. Bayes classifiers use probability theory and are fast, simple, and effective for text-based problems. Support Vector Machines find an optimal separating boundary and perform well in high-dimensional spaces. Each method has strengths and is chosen based on data size, complexity, and application needs.

52 Define classification.

Classification is a supervised learning task in which data items are assigned to predefined categories or classes based on their features. A classifier learns patterns from labeled training data and then predicts the class of new, unseen data. For example, an email can be classified as spam or non-spam, and an image can be classified as cat or dog. Classification is widely used in medical diagnosis, text categorization, image recognition, and fraud detection.

53 Explain how alpha–beta pruning helps reduce computation in fast-paced games

Alpha–beta pruning is an optimization technique used with the minimax algorithm to reduce the number of nodes evaluated in a game tree. It eliminates branches that cannot influence the final decision by maintaining two values: alpha (best value for maximizer) and beta (best value for minimizer). When beta becomes less than or equal to alpha, further exploration is stopped. This significantly reduces computation time and allows AI to make faster decisions in games like chess.

54 What is an intelligent agent?

An intelligent agent is an entity that perceives its environment through sensors and acts upon it through actuators to achieve goals. It selects actions based on percepts, internal knowledge, and reasoning mechanisms. Examples include a robot vacuum cleaner, a recommendation system, or a self-driving car. Intelligent agents aim to act rationally to maximize their performance measure.

55 Describe the working of hill climbing and simulated annealing algorithms.

Hill climbing is a local search algorithm that continuously moves to a neighboring state with better value until no improvement is possible. It is simple but can get stuck in local maxima. Simulated annealing improves hill climbing by occasionally allowing worse moves based on probability, helping escape local optima. It gradually reduces randomness over time, leading to near-optimal solutions. These algorithms are used in optimization and scheduling problems.

56 Explain A* algorithm with an example search tree.

The A* algorithm is an informed search algorithm, meaning it leverages a heuristic function to guide its search towards the goal. This heuristic function estimates the cost of reaching the goal from a given node, allowing the algorithm to prioritize promising paths and avoid exploring unnecessary ones.

The A* algorithm is a powerful and widely used graph traversal and path finding algorithm. It finds the shortest path between a starting node and a goal node in a weighted graph.

57 What is uninformed search?

Uninformed search is a search strategy that explores the search space without using any domain-specific knowledge. It relies only on the problem definition, such as initial state, actions, and goal test. Algorithms like Breadth-First Search (BFS) and Depth-First Search

(DFS) fall under this category. These methods are simple but inefficient for large search spaces.

58 What is heuristic seach?

Heuristic search uses problem-specific knowledge, called heuristics, to guide the search process toward the goal more efficiently. A heuristic estimates how close a state is to the goal. Algorithms like A* and Greedy Best-First Search use heuristics. Heuristic search significantly reduces search time compared to uninformed search.

59 Define the following problems. What types of control strategy is used in the following problem:

I. The Missionaries and cannibals problems II. 8-puzzle problem

A problem in AI consists of an initial state, goal state, actions, and a state space. A control strategy determines how the search proceeds. Both Missionaries and Cannibals and the 8-puzzle problem use **state space search** with **informed or uninformed control strategies**. BFS and A* are commonly applied. The control strategy ensures systematic exploration to reach the goal safely.
The Missionaries and Cannibals problem is a classic AI search problem where missionaries must cross a river without being outnumbered by cannibals.
The 8-puzzle problem involves arranging numbered tiles to reach a goal configuration using minimum moves. Both problems are modeled using state spaces and solved using search algorithms like BFS, DFS, or A*.

60 What is computer vision?

Computer Vision is the field of AI that enables machines to understand and interpret visual information from images and videos. It involves tasks such as image processing, feature extraction, object detection, and recognition. Computer Vision is used in facial recognition, medical imaging, surveillance, and autonomous vehicles.

61 What is uninformed search?

Uninformed search is a search strategy that does not use any domain-specific knowledge or heuristics. It explores the state space blindly based only on the problem definition such as initial state, goal test, and actions. Algorithms like Breadth-First Search (BFS) and Depth-First Search (DFS) are examples. Although uninformed search is simple and complete, it is inefficient for large or complex problems.

62 Define FOL (First Order Logic).

First Order Logic (FOL) is a knowledge representation language that extends propositional logic by allowing variables, predicates, and quantifiers. It can represent relationships between objects and express statements like "All humans are mortal." FOL is more expressive and powerful than propositional logic and is widely used in AI for reasoning and inference in expert systems.

62 What is a decision tree?

A decision tree is a supervised learning model used for classification and regression. It represents decisions using a tree-like structure where internal nodes denote conditions on features, branches represent outcomes, and leaf nodes represent class labels or values. Decision trees are easy to understand, interpret, and widely used in medical diagnosis and business decision-making.

63 Define feature extraction.

Feature extraction is the process of transforming raw data into meaningful and relevant features that can be used by machine learning algorithms. It reduces complexity by selecting important information and discarding irrelevant data. For example, edges and textures are extracted from images, and keywords are extracted from text. Proper feature extraction improves accuracy and efficiency.

64 What is the purpose of Bayesian reasoning?

Bayesian reasoning is used to make decisions under uncertainty by applying probability theory. It combines prior knowledge with new evidence using Bayes' theorem. This approach allows AI systems to update beliefs dynamically as new data is observed. Bayesian reasoning is commonly used in medical diagnosis, spam detection, and risk analysis.

65 Explain the need for probabilistic reasoning.

Probabilistic reasoning is needed because real-world information is often incomplete, uncertain, or noisy. Deterministic approaches fail in such environments. Probability allows AI systems to handle uncertainty and make rational decisions by estimating likelihoods. It improves prediction accuracy in applications like weather forecasting and disease diagnosis.

66 Explain how Bayesian networks help forecast weather by combining multiple uncertain variables.

Bayesian networks model weather variables such as temperature, humidity, air pressure, and rainfall as nodes connected by probabilistic relationships. These dependencies help the system combine uncertain observations effectively. When new data is observed, probabilities are updated to forecast outcomes like rain or storms. This approach provides accurate predictions despite uncertainty.

67 Explain how A* helps Google Maps find the fastest route considering: Distance, Time and Traffic congestion.

Google Maps uses A* search by considering the actual distance traveled and a heuristic estimate of remaining distance or time. Traffic conditions are incorporated into the cost function, making $g(n)$ dynamic. The heuristic $h(n)$ estimates remaining travel time. By minimizing $f(n)=g(n)+h(n)$, A* efficiently finds the fastest route in real time.

68 Explain the architecture and design principles of a pattern recognition system.

A pattern recognition system consists of data acquisition, preprocessing, feature extraction, classification, and decision-making stages. Preprocessing reduces noise, feature extraction highlights important patterns, and classifiers assign labels. Good design principles include robustness, accuracy, scalability, and generalization. These systems are used in speech recognition, biometrics, and image analysis.

69 Explain probabilistic reasoning in uncertain environments.

Probabilistic reasoning is an important technique in Artificial Intelligence used to handle uncertainty in real-world environments. In practical situations, information is often incomplete, noisy, or ambiguous, making it impossible to take decisions using strict logical rules alone. Probabilistic reasoning uses probability theory to model uncertainty and express how likely different outcomes are. It allows AI systems to make rational decisions based on likelihood rather than certainty.

In probabilistic reasoning, beliefs about the world are represented as probabilities. When new evidence is observed, these probabilities are updated using rules such as Bayes' theorem. Models like Bayesian networks, Markov models, and Hidden Markov Models are commonly used. For example, in medical diagnosis, symptoms do not guarantee a disease, but probabilistic reasoning helps estimate disease likelihood based on observed symptoms.

This approach improves decision-making because it allows AI systems to revise beliefs dynamically. Autonomous robots, weather forecasting systems, and spam filters all rely on probabilistic reasoning. Despite uncertainty, the system can choose actions that maximize expected benefit. Thus, probabilistic reasoning plays a crucial role in intelligent decision-making under uncertainty.

Bayesian Network showing nodes like *Cause → Effect → Evidence*

70 Discuss types of features used in pattern recognition.

Features are measurable characteristics extracted from raw data that help differentiate between classes in pattern recognition systems. The quality of features directly affects the system's accuracy and performance. Broadly, features are classified into statistical, structural, and transform-based features.

Statistical features describe numerical properties such as mean, variance, and texture. These features capture global patterns and are widely used because they are simple and computationally efficient. Structural features describe shape, topology, and spatial relationships, such as edges, corners, and contours in image recognition. They are useful in character and object recognition.

Transform-based features are obtained using mathematical transformations like Fourier Transform or Wavelet Transform. These features capture frequency and time information and are useful in speech and image processing. Selecting suitable features helps reduce dimensionality, noise, and computational cost. Effective feature selection improves classification accuracy and enhances system robustness.

Block diagram showing *Input → Feature Extraction → Classifier*

71 Explain the history of Artificial Intelligence.

Artificial Intelligence originated in the 1950s with the goal of creating machines that mimic human intelligence. Alan Turing laid the foundation by proposing the Turing Test. The term “Artificial Intelligence” was coined by John McCarthy in 1956 during the Dartmouth Conference. Early AI systems focused on symbolic reasoning, logic, and rule-based problem solving.

During the 1970s and 1980s, AI faced limitations due to insufficient computational power and data, leading to periods known as AI winters. Expert systems revived interest by encoding human knowledge into rule-based systems. However, they were expensive and hard to maintain.

The modern AI era began with machine learning and statistical methods, followed by deep learning in the 2010s. Advances in big data, GPUs, and neural networks led to breakthroughs in NLP, computer vision, and speech recognition. Today, AI is widely applied across healthcare, finance, robotics, and education.

72 Explain different types of intelligent agents with diagrams and examples.

An intelligent agent is a system that perceives its environment using sensors and acts upon it using actuators to achieve goals. Intelligent agents are classified based on their decision-making abilities.

A **simple reflex agent** acts only on the current percept using condition–action rules, such as a thermostat. A **model-based agent** maintains an internal model of the environment, enabling better decisions in partially observable environments, such as a robot vacuum. A **goal-based agent** selects actions to achieve specific goals, such as navigation systems. A **utility-based agent** chooses actions by maximizing utility, for example, stock trading systems. A **learning agent** improves performance through experience, such as spam filters.

These agents are used depending on task complexity and environment uncertainty.

Agent–Environment diagram (Sensors → Agent → Actuators)

73 Explain dimensionality reduction.

Dimensionality reduction is the process of reducing the number of features in a dataset while preserving essential information. High-dimensional data increases computation, noise, and risk of overfitting, known as the curse of dimensionality. Dimensionality reduction simplifies models and improves performance.

Techniques like Principal Component Analysis (PCA) transform data into a smaller set of uncorrelated variables. Linear Discriminant Analysis (LDA) focuses on maximizing class separability. Feature selection methods choose the most relevant features directly. These methods help improve accuracy, reduce training time, and enhance visualization.

Dimensionality reduction is widely used in image processing, bioinformatics, and text mining, making AI models more efficient and robust.

74 Discuss Natural Language Processing and explain its major components.

Natural Language Processing (NLP) enables computers to understand, interpret, and generate human language. NLP bridges the gap between human communication and machine understanding using linguistic and statistical techniques.

The major components of NLP include lexical analysis (tokenization and morphology), syntactic analysis (parsing sentence structure), semantic analysis (meaning representation), discourse processing (context understanding), and pragmatic analysis (intent understanding). For example, chatbots use NLP to understand user queries and generate responses.

NLP applications include machine translation, speech recognition, sentiment analysis, and information retrieval. With deep learning models like transformers, NLP systems have become more accurate and context-aware.

Pipeline: *Text → Tokenization → Parsing → Semantics → Output*

75 Explain K-means clustering algorithm.

K-means is an unsupervised learning algorithm used to group data into K clusters based on similarity. Initially, K cluster centroids are chosen randomly. Each data point is assigned to the nearest centroid, and then centroids are recalculated as the mean of assigned points. This process repeats until clusters stabilize.

K-means is simple, fast, and efficient for large datasets. It is used in customer segmentation, image compression, and document clustering. However, it requires the number of clusters to be specified in advance and is sensitive to initial centroid selection.

76 Compare BFS and DFS.

Breadth-First Search (BFS) explores nodes level by level, while Depth-First Search (DFS) explores one branch deeply before backtracking. BFS guarantees finding the shortest path in unweighted graphs but requires high memory. DFS is memory-efficient but may get stuck in deep infinite paths.

BFS uses a queue data structure, whereas DFS uses a stack or recursion. BFS is suitable for shallow search problems, while DFS works well for deep problems with limited memory. Both are uninformed search strategies.

77 Explain backward chaining and its application.

Backward chaining is a goal-driven reasoning technique where inference starts from the goal and works backward to find supporting facts. The system checks which rules can lead to the goal and verifies whether their premises are true.

Backward chaining is widely used in expert systems and diagnostic applications, such as medical diagnosis. It is efficient when the number of goals is small. Compared to forward chaining, it avoids unnecessary rule evaluations.

Goal ← Rules ← Facts

78 Write a short note on Naive Bayes classifier.

Naive Bayes classifier is a **probabilistic machine learning algorithm** based on **Bayes' Theorem**. It is called "naive" because it assumes that all features are **independent of each other**, which is not always true in real-life data. Despite this assumption, Naive Bayes performs very well in many applications. It is widely used in **spam email detection, sentiment analysis, and document classification**. The classifier calculates the probability of each class for a given input and assigns the input to the class with the highest probability. Naive Bayes is fast, easy to implement, and works well even with large datasets and limited training data.
Diagram idea: Features → Probability calculation → Class with maximum probability.

79 Explain minimax algorithm with a complete game tree.

The Minimax algorithm is an **adversarial search algorithm** used in two-player games such as chess, tic-tac-toe, and checkers. One player is called **MAX**, who tries to maximize the score, while the other is called **MIN**, who tries to minimize it. The algorithm constructs a **game tree** where nodes represent game states. Leaf nodes contain utility values. These values are propagated upward: MAX nodes select the highest value, and MIN nodes select the lowest value. Finally, the root node gets the best possible move. Minimax assumes that the opponent always plays optimally.
Diagram idea: Game tree showing MAX and MIN levels with utility values at leaves.

80 Discuss backward chaining with small example.

Backward chaining is a **goal-driven inference technique** mainly used in expert systems. In this approach, the system starts with a goal and works backward to determine which facts and rules can satisfy that goal. If a required fact is not known, it becomes a new sub-goal. The process continues until the goal is proven or rejected.

Example:

Rule: IF it is raining → ground is wet

Goal: ground is wet

System checks whether "it is raining" is true.

If true, the goal is satisfied.

Diagram : Goal → Rules → Sub-goals → Known facts.

81 Compare minimax and alpha-beta pruning.

Minimax and Alpha-Beta pruning are both algorithms used in game playing. Minimax explores **all possible moves** in the game tree, which makes it slow and computationally expensive. Alpha-Beta pruning improves Minimax by **eliminating branches** that cannot affect the final decision. Alpha represents the best value for MAX, and Beta represents the best value for MIN. When alpha becomes greater than or equal to beta, further exploration is stopped. Alpha-Beta pruning gives the **same optimal result as Minimax** but with much less computation.

82 Explain inference in first-order logic with examples of resolution.

Inference in First-Order Logic (FOL) is the process of deriving new facts from existing knowledge using logical rules. **Resolution** is a powerful inference technique where statements are first converted into clauses using **Conjunctive Normal Form (CNF)**. Clauses with complementary literals are combined to produce new clauses. If an empty clause is derived, the statement is proven.

Example:

$\forall x$ Human(x) → Mortal(x)

Human(Socrates)

Conclusion: Mortal(Socrates)

Diagram: Facts → Clauses → Resolution steps → Conclusion.

83 What is propositional inference?

Propositional inference is a reasoning process where new facts are derived using **propositional logic rules**. It uses logical symbols such as P, Q, and R, which can be either true or false. Common inference rules include **Modus Ponens, Modus Tollens, and Resolution**. Propositional inference is simple and widely used in automated reasoning systems, but it cannot represent complex relationships involving objects and properties.

84 Explain the use of local search algorithms in scheduling airline crew or production planning.

Local search algorithms such as **hill climbing and simulated annealing** are widely used in optimization problems like airline crew scheduling and production planning. In airline scheduling, the goal is to assign crew members efficiently while minimizing cost and meeting constraints. The algorithm starts with an initial solution and continuously improves it by moving to better neighboring solutions. In production planning, local search helps in optimal resource allocation. These methods are fast and suitable for large problem spaces.

85 What is local beam search?

Local Beam Search is an extension of hill climbing where instead of keeping one current state, the algorithm keeps **k best states** at each level. From these states, all successors are generated, and the best k are selected for the next step. This reduces the chance of getting stuck in a local maximum. Local beam search is used in optimization and scheduling problems.

86 Compare informed and uninformed search strategies with suitable examples.

Uninformed search does not use any heuristic information. Examples include **Breadth First Search (BFS)** and **Depth First Search (DFS)**. These algorithms blindly explore the search space. Informed search uses **heuristic functions** to guide the search toward the goal. Examples include **A*** and **Greedy Best-First Search**. Informed search is generally faster and more efficient.

87 Define admissible heuristic.

An admissible heuristic is a heuristic function that **never overestimates the actual cost** to reach the goal. It always gives an optimistic estimate. Admissible heuristics guarantee that algorithms like A* search will find an optimal solution.

88 What is the difference between local and global search?

Local search focuses on improving a single solution without keeping track of the entire search path. Examples include hill climbing. Global search explores the entire search space and maintains paths, such as BFS. Local search is memory efficient, while global search guarantees finding a solution if one exists.

89 What is Bayesian network?

A Bayesian Network is a **probabilistic graphical model** that represents random variables and their conditional dependencies using a **Directed Acyclic Graph (DAG)**. Nodes represent variables, and edges show dependence. Bayesian networks are useful for reasoning under uncertainty, such as medical diagnosis and weather prediction.

90 Define first-order logic.

First-Order Logic (FOL) is an extension of propositional logic that includes **objects, predicates, variables, and quantifiers** such as ∀ (for all) and ∃ (there exists). It can represent complex facts about the real world.
Example: $\forall x\ Human(x) \rightarrow Mortal(x)$

91 What is inference?

Inference is the process of drawing **logical conclusions** from known facts and rules. It is a core concept in AI and is used in reasoning, problem-solving, and decision-making systems.

92 Define Hidden Markov Model.

A Hidden Markov Model is a **probabilistic model** where system states are hidden, but outputs are observable. It uses transition probabilities and emission probabilities. HMMs are widely used in **speech recognition, handwriting recognition, and bioinformatics**.

93 What is informed search?

Informed search is a search strategy that uses **heuristic information** to guide the search toward the goal efficiently. Algorithms like A* search and greedy search are informed searches. They reduce time and space complexity.

94 Define the term "state space."

The state space of a problem is the **set of all possible states** that can be reached from the initial state, including the goal state. It helps in structuring and solving AI search problems. **Diagram idea:** Initial state → Intermediate states → Goal state.

95 What is hill climbing?

Hill climbing is a **local search algorithm** that continuously moves toward a better neighboring state. It is simple and fast but may get stuck at local maxima where no better neighboring state exists.

96 Define inference in logic.

Inference in logic is the method of applying **logical rules** to existing statements to derive valid conclusions. It forms the backbone of logical reasoning in AI systems.

97 Differentiate between propositional logic and first-order logic.

Propositional logic deals only with simple true or false statements, while First-Order Logic handles **objects, relationships, and quantifiers**. First-Order Logic is more expressive and powerful than propositional logic.